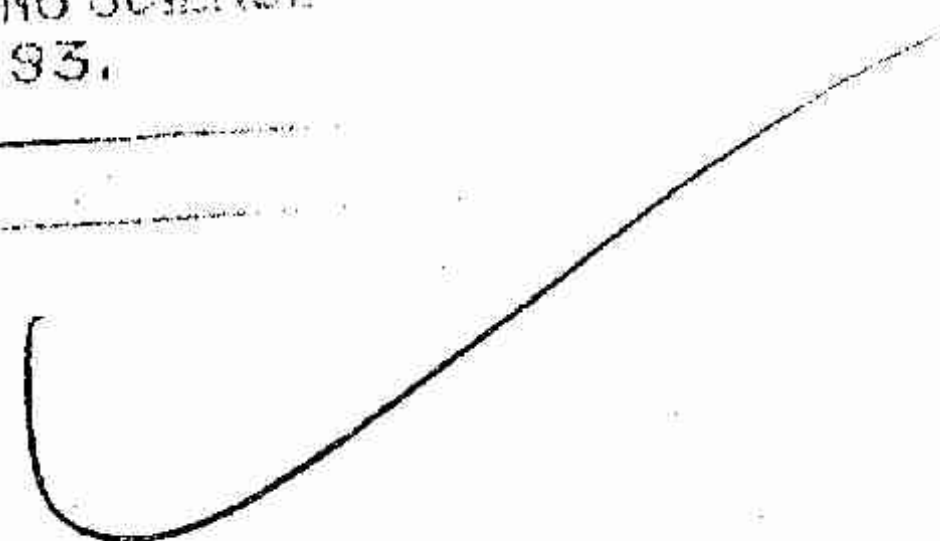


12

ZIMBABWE SCHOOL EXAMNATIONS COUNCIL
General Certificate of Education Advanced Level

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MARKING SCHEME

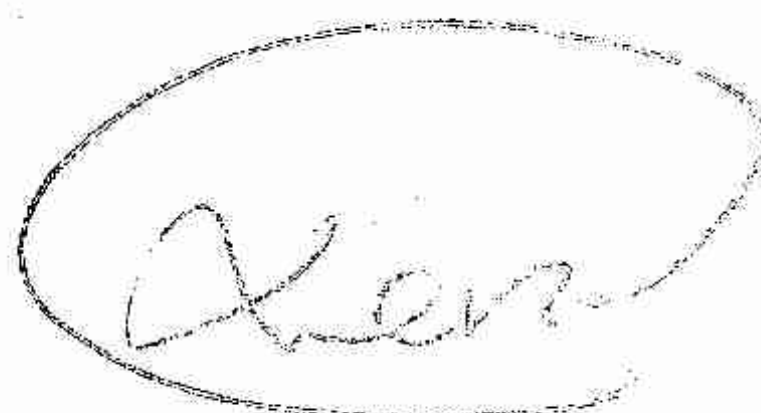
JUNE 2014

BIOLOGY



9190/2

7, 8, 9, 12, 13



- 1 (a) (i) Q, [1]
 (ii) it has a capsule/slime layer; [1]

(b)

Cell Q	Cell R	Cell S
x	✓	x;
✓	✓	✓;

[2]

NB: Treat blank as cross if in the absence of other crosses in table;

- (c) (i) - Bacterial cell wall has no cellulose/plant cell wall has cellulose for support and expulsion of excess water; [1]
 (ii) - water potential is lower/more negative inside cell; [2]
 - water enters cell by osmosis/diffusion;
 (d) - plasma membrane is thin/~~small~~; 7nm
 - electron microscope has greater resolution/wavelength of ~~electrons~~ ^{electrons} short; [2]

[Total: 9]

- 2 (a) (i) activation energy/amount of energy required for reaction ^{for rxn to start}; [1]
 (ii) curve starting and finishing at correct energy levels; [2]
 activation energy lower (i.e. less than x);
 (iii) energy in products less (than in substrate/hydrogen peroxide); [2]
 energy given off/lost as heat/exergonic/exothermic;

- (b) - (molecules have) less (kinetic) energy;
 - move slower;
 - fewer collisions/fewer enzyme-substrate complexes formed;

~~max~~ 3 max
 [max 2]
 [Total: 7]

- (a) (i) - 2 cells/2 nuclei;
 - same chromosome number/preserves chromosomes number/
 $n \rightarrow n/2n \rightarrow 2n$;
 - genetically identical/same DNA;
 - identical to parent/~~identical to each other~~;

(4)
 [max 2]

- (ii) - growth; @ NOT growth of cells
 - repair;
 - replacement of cells;
 - named e.g;

4
 [max 2]

- (b) @ stress, pollution, diet
 - radiation/mobile phones/radon;
 - UV light/sunlight/sunburn;
 - X-rays;
 - gamma rays;
 - smoking/tobacco;
 - named carcinogen (tar, aniline dyes, asbestos, alcohol, benzyopyrene)
 benzene, vinyl chloride)
 - (named) virus; **DNA/RNA**
 - heredity/AW;
 - low fibre diet;
 - high fat diet;
 - free radicals;
 - increased age;
 - weakened immunity;
 - high voltage – power cables;

14
 max
 [max 2]
 [Total: 6]

- 4 (a) - peptide bonds between amino acids;
 - primary structure/amino acid sequence determines folding ~~side~~ **side**;
 - hydrogen bonds maintain (secondary structure);
 - ref. beta pleated sheet/alpha helix;
 - folding to form tertiary structure/globular shape;
 - sulphur bridges/ionic bonds/van der Waals forces/ hydrophobic interactions;

[max 3]

- (b) - reverse transcriptase/synthesise DNA from mRNA;
 - restriction endonuclease/enzyme;
 - produces sticky ends;
 - plasmid cut by same restriction enzyme;
 - recombinant DNA formed;
 - DNA ligase; ref;
 - DNA inserted into host;
 - detail of insertion **by** electric shock/calcium ions;

8 max 3
~~[max 4]~~

- (c) - no contamination/ref named infection/HIV/reduced rate of infection/
greater production rate; [1]
- (d) - ref. to introns and exons/only mammalian cells have Golgi/enzymes
(for post translational modification); [1]
- [Total: 8]

- 5 (a) cytoplasm; [1]
- (b) fructose (1,6) biphosphate/phosphorylated 6-carbon sugar; [1]
- (c) - hexose energy rich;
- does not react easily/AW;
- phosphorylation activates hexose;
- maintains concentration gradient of glucose inside and outside cell; 4 max [2]
- (d) - ~~1~~ does not enter Krebs cycle;
- decarboxylated/ CO_2 released;
- forms ethanal;
- to ethanol;
- reduced NAD does not enter ETC;
- ref. alcohol dehydrogenase;
- ~~ref~~ irreversible; 7 max [5]
- [Total: 9]

- 6 (a) (i) A - sodium;
B - potassium; [2 max]
- (ii) - sodium ions diffuse in;
- causing depolarisation/described;
- potassium ions then diffuse out, to start recovery; 3 max [2 max]
- (b) - active transport/pumping of sodium (ions across membrane);
- out of neurone/higher concentration outside;
- differential permeability to K^+ and Na^+ ; 3 [2 max]
- (c) no (new) action potential/nerve impulse produced in this time; [1]
- (d) - speed up transmission of ~~transmission~~ nerve impulse;
- by insulating the axon membrane; *Insulation only*
- sodium and potassium ions cannot flow through the myelin sheath;
- only occur at nodes of Ranvier; *saltatory conduction* [3 max 2]
- [Total: 9]

- (a) A - nitrogen fixation;
 B - nitrification;
 C - denitrification; [3]

- (b) - eutrophication;
 - death of aquatic organisms;
 - soil becomes acidic;
 - A.V.P; [max 3]

- (c) - legumes/green manure/compost/organic matter/humus/manure/ploughed/
 dug into soil;
 - grew legumes for nitrogen fixation;
 - ref. improve aeration/drainage; [max 2]
 [Total: 8]

- 8 (a) A - frond;
 B - rachis;
 C - ~~rhizome~~, root. [3]

- (b) - filicinophyta; [1]

- (c) - presence of true roots; stem and leaves;
 - presence of vascular tissue;
 - sporophyte generation dominant/no longer dependent on gametophyte; [3]
 - ~~ramenta~~ ~~stomata~~ - reduce dessication. [Total: 7]
 - ~~stomata~~ for gaseous exchange
 - AVP

- 9 (a) (i) LH peaks/is high (on day 12); [1]

- (ii) sperm can survive up to 2 - 7 days;
 eggs can survive for hours/up to 3 days;
~~or~~/general idea of short-lived gametes (e.g. a few days), (for both points).
 ovulation would have occurred/egg present; [2]

- (b) (i) - inhibits (release of) FSH;
 - Follicle does not develop/mature/grow/prevent ovulation/
~~FSH causes follicles to develop~~ ~~A~~ ovum not produced; [2]
 [Total: 5]

10

- (a) 1 Amino acid based on carbon with four groups attached;
 2 amino/ NH_2 and carboxyl/ COOH ;
 3 R group/side chain + hydrogen;
 4 R-group differs from one amino acid to another;
 5 amino acids joined by condensation;
 6 bond formed between NH_2 and COOH ;
 7 involves removal of molecule of water;
 8 H from NH_2 and OH from COOH ;
 9 forming peptide bond;

Accept an
annotated
diagram

9
max
[8]

- (b) 10 Polymer/polysaccharide/AW;

11 2 components ~~amylose~~ and amylopectin

12 glucose;
 13 detail of structure on (atom 1; C-OH down) / - diagram of a glucose;

14 condensation;

15 glycosidic bond;

16 joined by 1,4 linkages;

17 compact;

18 amylose = non-branching/straight/AW;

19 coils/is a helix;

20 amylopectin = branching form;

21 1,6 links;

22 more/up to twice the number, glucose (residues) / amylose 1:2 amylopectin;

14
- [max 8]

[Total: 16]

11

- (a) 1 thick elastic layer in artery;

2 even outflow/associated with recoil;

3 ~~link between pressure in artery and ventricle contraction/systole~~ (3) thick muscular layer;

4 ~~arteriole with muscular layer~~ (4) to allow peristaltic flow of blood;

5 ~~muscle contraction, results in smaller diameter/vasoconstriction~~ endothelium layer;

6 ~~alter blood supply to different organs~~ smooth flow of blood;

7 ~~endothelium provides smooth surface/limits friction~~ (thick) collagen layer (tough)

8 ~~capillary wall thin/only endothelium~~ (8) small lumen (8) to allow rapid flow

9 ~~ret to gaps~~ (9) capillary from a branched network;

10 ~~for exchange of materials; white blood cells~~ (10) narrow lumen (10) squeezed flat & diffusion more efficient;

- (b) 20 At arteriole end of capillary;

21 hydrostatic pressure;

22 forces out soluble/small molecule;

23 and water;

24 protein remains in blood/plasma;

25 molecules too large;

26 more negative water potential at (venule) end;

27 water drawn in by osmosis/diffuses in;

28 some fluid returned (to blood) by lymphatic system;

9
[max 8]

[Total: 16]

12

- (a)
- 1 Individuals in population have great reproductive potential/AW;
 - 2 numbers in population remain roughly constant;
 - 3 may fail to survive/die;
 - 4 do not reproduce;
 - 5 due to environmental factors/named factors;
 - 6 variation in members of population;
 - 7 those best adapted survive;
 - 8 reproduce/pass on alleles; @ genes; *traits / x'ties*
 - 9 genetic variation leads to change in phenotype;
 - 10 ref. changes in gene pool;
 - 11 overtime produces evolutionary changes;
 - 12 new species arise from existing ones;

12
max
[max 8]

- (b)
- 13 (gene) example; (sickle cell anaemia/phenylketonuria/PKU)
 - 14 change in gene/DNA/base change;
 - 15 different amino acid;
 - 16 different polypeptide/different protein/non-functional protein;
 - 17 AVP; details *of specific example*

17-20

- 13 (chromosome) e.g. (Down's/Turner's syndromes)
- 14 structural changes in chromosomes;
- 15 change in number of chromosome;
- 16 change in sets of chromosomes;

- ~~change in sets of chromosomes/ref. polyploidy;~~
- 17 AVP; details *-20 of specific example*

[max 8]
[Total: 16]

13

- (a) 1 two from

1 (i) cuttings;

- 2 root/shoot/stem/leaf/branch, cut;
- 3 placed on, moist medium/compost/soil;
- 4 PGR's/named e.g. A.A./rooting hormone applied;
- 5 mitosis/division of cambium/callus cells;

6 (ii) layering;

- 7 bend runner horizontally;
- 8 cut notch through base of stem/remove lower leaves;
- 9 detach from parent when roots form;
- 10 roots/new shoots, form at axial bud;

11 (iii) grafting/budding;

- 12 root with vigorous growth/healthy roots;
- 13 stock;
- 14 shoot/bud, chosen with desirable, flowers/fruit;
- 15 scion;
- 16 match slanting cuts/AW;
- 17 bind together;
- 18 cell division/mitosis;
- 19 vascular tissue/xylem/phloem, fuses;

max of 4
marks from
each method.

Totipotency - ability of a cell to divide & produce all of the differentiated cells in an organism